NOTE

4-connected Maximal Planar Graphs are 4-ordered

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Dedicated to Danny Kleitman with thanks.

Abstract

It is shown that in a 4-connected maximal planar graph there is for any four vertices a, b, c and d, a cycle in the graph that contains the four vertices and visits them in the order a, b, c and d.

1 Introduction

All our graphs are simple (i.e., without loops or multiple edges). A graph is said to be k**-cyclable** if given any set of k vertices there is a cycle that contains the k vertices. A cycle cannot have repeated vertices. (We use the notation and terminology of [2].) This property was introduced by Watkins and Mesner [9], who characterised 3-cyclable graphs and gave sufficient conditions for a graph to be k-cyclable. It is well known that being 2-connected is equivalent to being 2-cyclable, and that in general being k-connected implies k-cyclable. Also, a hamiltonian graph is one that is k-cyclable for all k.

Recently, Ng and Schultz [6] introduced the concept of orderability. Following [3], we say that a graph is k**-ordered** if given any set of k vertices there is a cycle through the k vertices in any specified order. Being 3-ordered is equivalent to being 3-cyclable, but for $k \geq 4$ being k-ordered is stronger than being k-cyclable. In fact, it is easy to show that being k-ordered implies being $(k-1)$-connected [6]. For vertices $a_1, a_2, \ldots, a_m$ we will use the term $(a_1, a_2, \ldots, a_m)$-cycle to mean a cycle that visits these vertices in the order specified.

We are interested here in planar graphs. The first result in this direction was proved by Sallee [7] who showed that a 3-connected planar graph is 5-cyclable. This is best possible as there are 3-connected maximal planar graphs that are not 6-cyclable. (Such graphs were characterised by Kelmans and Lomonosov [5].) Furthermore, a 4-connected planar graph is hamiltonian and hence k-cyclable for all k.

Turning to k-ordered planar graphs, we start with a few simple observations:

Lemma. *Let G be a planar graph.*
1) If G is 3-connected then G is 3-ordered.
2) If G is 4-ordered, then G is maximal planar, 3-connected, and any cut-set of cardinality 3 isolates a vertex.
3) G is not 5-ordered.

PROOF. 1) Follows from Sallee [7], for example.

2) Assume G is 4-ordered. If there is a face of G which is not a triangle, then let a, b, c and d be consecutive vertices on the face. Then there is no (a,c,b,d)-cycle (by Jordán curve theorem).

If there is a cut-set of cardinality 3 that produces two nontrivial components, then let a, b be two vertices on one side of the cut, and c and d two vertices on the other side. There is no (a,c,b,d)-cycle.

3) We know that being 5-ordered requires being 4-connected. Consider any vertex v and let a, b, c and d be four of its neighbours in order in a plane embedding of G. Then there is no (a,c,v,b,d)-cycle. QED

So the problem remains of when is a planar graph 4-ordered. Our theorem is that the necessary conditions given in 2 above are almost sufficient:

Theorem. *A 4-connected maximal planar graph is 4-ordered.*

We prove this result in the next section. It uses a lengthy but standard inductive argument.

There is also a connection with 2-linked graphs. A graph is said to be **2-linked** if for any four vertices a, b, c and d there are disjoint paths from a to b and from c to d. Clearly, being 4-ordered is a stronger condition than being 2-linked. Jung [4] showed that a 4-connected graph is 2-linked unless it is a planar graph that is not maximal. As a corollary it follows that being 6-connected implies being 2-linked. Thomassen [8] characterised when graphs are 2-linked.

Faudree [3] asked whether being 6-connected implies being 4-ordered. This problem remains open. Since being 4-linked implies being 4-ordered, there is at least some value of connectivity that implies a graph is 4-ordered by the result of [1].

2 Proof of Theorem

The proof is by induction. The base case is K_4.

Let G be a 4-connected maximal planar graph. Let $A = \{a,b,c,d\}$ and assume we need an (a,b,c,d)-cycle. We say two vertices of A are **consecutive in *A*** if they must appear consecutively in the desired cycle. (For example, a and b are consecutive in A, but a and c are not.) We denote the neighbourhood of a vertex v by $N(v)$. If e is any edge, we denote by $G \cdot e$ the simple graph formed by contracting e to a single vertex.

Claim 1 *For any edge e, the graph $G \cdot e$ is maximal planar. Furthermore, if $G \cdot e$ is not 4-connected, then e is in an induced cut-4-cycle.*

PROOF. Consider any edge $e = uv$. Since G is 4-connected and maximal, the vertices u and v have exactly two common neighbours. Therefore the graph $G \cdot e$, formed by contracting u and v to a single vertex x, is maximal planar.

Suppose $G \cdot e$ is not 4-connected. Then it has a cut-set B of cardinality 3 with $x \in B$. It follows that $B' = B - \{x\} \cup \{u, v\}$ separates G. Since B' is a minimal cut-set, the subgraph induced by B' is isomorphic to an induced cycle, and contains e. QED

Claim 2 *We may assume that: Every edge not incident with A lies in an induced cut-4-cycle.*

PROOF. Suppose edge $e = uv$ is not incident with A and that $G \cdot e$ formed by contracting u and v to vertex x is 4-connected. Then, by the inductive hypothesis, in $G \cdot e$ there is an (a, b, c, d)-cycle C. This extends to an (a, b, c, d)-cycle in G as follows. If C avoids x, then it is a cycle in G. If C uses x, then let s and t be the vertices either side of x in C. Then at least one of su or sv is an edge in G, as is at least one of tu or tv. By replacing sx and xt with these two edges, together with e if necessary, we obtain the (a, b, c, d)-cycle in G. Hence we may assume that $G \cdot e$ is not 4-connected, and thus by Claim 1, e lies in an induced cut-4-cycle. QED

Claim 3 *We may assume that: Every vertex not in A has degree at least* 5.

PROOF. Suppose vertex $u \notin A$ has degree 4. Consider any edge $e = uv$ incident with u. We already know that if $v \notin A$ then the edge e is in an induced 4-cycle (Claim 2).

But this is true even if $v \in A$. For, suppose that $G \cdot e$ is 4-connected. Then, by the inductive hypothesis, there is in $G \cdot e$ a cycle C through A in the desired order where x takes the place of v. This extends to an (a, b, c, d)-cycle in G as follows. All but one neighbour of x is also a neighbour of v; thus C uses an edge sx where s is a neighbour of v. Hence we can convert C to the desired cycle.

So we may assume that every edge incident with u is in an induced cut-4-cycle. Let the neighbours of u be v_1, v_2, v_3, v_4 in order in the embedding of G. The induced cut-4-cycle for edge uv_1 contains v_3 and some other vertex, say y. The induced cut-4-cycle for edge uv_2 contains v_4 and some other vertex, say z. By planarity, $z = y$. In particular, the subgraph induced by the six vertices $N(u) \cup \{u, y\}$ is a triangulation, and therefore the whole graph is the octahedron. It is easily checked that G is 4-ordered, and we are done. QED

Claim 4 *We may assume that: If D is a cut-4-cycle of G, then on both sides of D there is at least one vertex of A. Furthermore, if on one side there is only one vertex of A, then it is the only vertex on that side. Also, if on one side there are exactly two vertices of A, then they are not consecutive in A.*

PROOF. Let X be the set of vertices inside D. Let G' be the (maximal planar) graph formed by contracting X to a single vertex x. Then G' is still 4-connected. (Note that by planarity, for any two vertices $u, v \notin X$, if not both on D then at most one minimal u–v path in G can use X and so there are still four internally disjoint u–v paths in G'. The case when u and v both on D is easily checked.)

Suppose $|X \cap A| = 0$. If $|X| = 1$ then the vertex of X has degree 4, contradicting the above claim. So G' is smaller than G and we can apply the inductive hypothesis to G' to obtain an (a, b, c, d)-cycle in G' which corresponds to an (a, b, c, d)-cycle in G, and we are done.

Suppose $|X \cap A| = 1$ with $a \in X$. If G' has fewer vertices than G, then we can again apply the inductive hypothesis (with x substituting for a) and uncontract to obtain the desired (a, b, c, d)-cycle. Hence, we may assume that $|X| = 1$, as required.

Suppose $|X \cap A| = 2$ with $a, b \in X$. By 4-connectivity, there are four internally disjoint paths from a to the four vertices of D. We claim we can choose the paths such that one contains b. For if not, then consider the four internally disjoint paths from b to D: two of these intersect the same a-to-D path, and hence b can be spliced into that path. Say the a-to-D path containing b ends at vertex u.

By induction, we can find an (x, u, c, d)-cycle C in G'. Say the two edges of C incident with x are vx and xu. By replacing these by the v-to-a and a-to-u-via-b paths constructed above, we obtain the desired (a, b, c, d)-cycle. QED

Now, let Q be the set of vertices outside A of degree exactly 5. By Euler's formula and Claim 3, since the vertices of A have degree at least 4, it follows that $|Q| \geq 4$. Recall that an embedding of a graph on a surface is equivalent to a cyclic ordering of the edges at each vertex. Thus we may speak of **successive neighbours of a vertex**.

Claim 5 *We may assume that: If $u \in Q$, then exactly two of its neighbours, say x and y, are in A. These two neighbours are not successive neighbours of u in the planar embedding of G, but are consecutive in A. Further, neither of the edges ux and uy is in a cut-4-cycle.*

Proof. Let $u \in Q$, and let its neighbours be $v_1, v_2, \ldots, v_5$ in order. Suppose first that three successive neighbours of u are not in A: say v_1, v_2, v_3. The cut-4-cycle for uv_2 uses another neighbour of u, say v_4, and an external vertex y. Then consider the cut-4-cycle for uv_3. It must use the same external vertex y, by planarity. But then the four regions around v_3 are triangles, and so v_3 has degree 4, a contradiction of Claim 3.

Suppose three successive neighbours of u are in A: say v_1, v_2, v_3. Then let d be the remaining vertex of A. By connectivity, there are four internally disjoint paths from d to the set $\{u, v_1, v_2, v_3\}$ and from this the desired cycle follows.

So we may assume that for any three successive neighbours of u, at least one is in A and at least one is out of A. By symmetry, we may assume that $v_1, v_3 \notin A$, and that $v_2, v_4 \in A$. By Claim 4, uv_2 cannot be in a cut-4-cycle (since there would be a non-A-vertex on both sides of the cycle and hence at least two A-vertices on both sides, contradicting the cardinality of A). So $G \cdot uv_2$ is 4-connected.

If v_2 and v_4 are not consecutive in A (e.g. $v_2 = a$ and $v_4 = c$), then we may contract uv_2 to x and induct to obtain an (x, b, c, d)-cycle C in $G \cdot uv_2$. Since u has degree 5 and C cannot use the edge xv_4, at least one of the edges incident

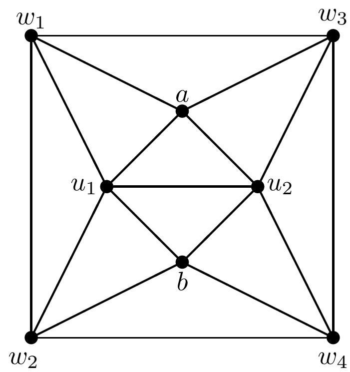

Figure 1: A subgraph of G

with x in C is incident with a neighbour of v_2; so we obtain the desired cycle in G. Hence we may assume that v_2 and v_4 are consecutive in A.

Finally, suppose that $v_5 \in A$. Then, by the above, v_2 and v_5 are consecutive in A. Say $v_2 = b$, $v_5 = a$ and $v_4 = c$. By 4-connectivity, there are four internally disjoint paths from d to the set $N(u)$. If both a and c are endpoints of such paths we are done. So suppose a is not. Then we construct the desired cycle by using the path from d to v_1, then the path $v_1, v_5, u, v_2, v_3, v_4$ and the path from v_4 back to d. Hence we may assume that $v_5 \notin A$. QED

If there is a separate vertex of Q for each of the four pairs of consecutives in A, then we are done (as we obtain an (a, b, c, d)-cycle of length 8). So since $|Q| \geq 4$ we may assume that there is a consecutive pair of A, say a, b, such that two vertices of Q are involved, say u_1 and u_2.

Consider the 4-cycle $D : a, u_1, b, u_2, a$. If this is a cut-cycle, then by Claim 4 there is at least one vertex of A both in- and outside of D, and at least two on one side, a contradiction. So there is no vertex on one side of D, say the inside. Since a and b are not successive neighbours of u_1, there is an edge from u_1 to u_2 inside D. By the lack of cut-triangles it follows that $N(u_1) \cap N(u_2) = \{a, b\}$. Say the neighbours of u_1 are in clockwise order a, u_2, b, w_2, w_1 and those of u_2 are b, u_1, a, w_3, w_4. Let $W = \{w_1, w_2, w_3, w_4\}$.

Now, the cut-4-cycle for edge $w_1 u_1$ that is guaranteed by Claim 2 must use u_2 (by Claim 5) and hence one of w_3, w_4. That is, w_1 is adjacent to one of w_3 or w_4. Similarly, w_2 is adjacent to one of w_3 or w_4. Further, both w_3 and w_4 are adjacent to one of w_1 or w_2. From this and planarity it follows that both $w_1 w_3$ and $w_2 w_4$ are edges. Hence we have the subgraph shown in Figure 1.

Finally, we observe that w_1, w_2, w_4, w_3, w_1 is a cut-4-cycle, and it contradicts Claim 4.

3 Open Question

A possible extension is to ask for a subdivision of K_4 with a, b, c, d as the four vertices of the K_4. This is not always possible. It remains an open question if one can find a subdivision of $K(2,3)$ where either a and c or b and d are the vertices of degree 3.

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References

[1] B. Bollobás and A. Thomason, Highly linked graphs, *Combin. Prob. Comput.* **2** (1993), 1–7.

[2] G. Chartrand and L. Lesniak, *Graphs & Digraphs (3rd ed.)*, Chapman & Hall, 1996.

[3] R.J. Faudree, Survey on results on k-ordered graphs, *Discrete Math.* **229** (2001), 73–87.

[4] H.A. Jung, Eine verallgemeinerung des n-fachen zusammenhangs für Graphen, *Math. Ann.* **187** (1970), 95–103.

[5] A.K. Kelmans and M.V. Lomonosov, When m vertices in a k-connected graph cannot be walked round along a simple cycle, *Discrete Math.* **38** (1982), 317–322.

[6] L. Ng and M. Schultz, k-Ordered hamiltonian graphs, *J. Graph Theory* **24** (1997), 45–57.

[7] G.T. Sallee, Circuits and paths through specified nodes, *J. Combin. Theory Ser. B* **15** (1973), 32–39.

[8] C. Thomassen, 2-linked graphs, *Europ. J. Combin.* (1980), 371–378.

[9] M.E. Watkins and D.M. Mesner, Cycles and connectivity in graphs, *Canad. J. Math.* **19** (1967), 1319–1328.